

Equation of motion

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\} + \delta x_D[k],$$

Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{f_{dx}[k] + f_T[k]\} - \delta x_D[k],$$

$\delta x[k] = x_d[k] - x[k]$, $x_d[k]$ is the desired motion trajectory, $\delta x_D[k]$ is the motion error caused by the disturbance, and $\Delta x_d[k] = x_d[k+1] - x_d[k]$ is the desired motion step size.

Measurement

(d -step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k],$$

Control objective: $\delta x[k+1] = \lambda_c \cdot \delta x[k]; \quad 0 \leq \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dx}[k] = a_x^{-1}[k]\{\Delta x_d[k] + (1 - \lambda_c) \delta \mathbf{x}[\mathbf{k}] - \delta x_D[k]\}$$

Motion error:

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{f}_{dx}[k] + f_T[k]\} - \delta x_D[k] = \lambda_c \cdot \delta x[k] - a_x[k]f_T[k]$$

d -step measurement delay:

$$\delta \mathbf{x}[\mathbf{k}] = \delta x[k-d] + \sum_{i=1}^d \Delta x_d[k-i] - \sum_{i=1}^d a_x[k-i]\{f_{dx}[k-i] + f_T[k-i]\} - \sum_{i=1}^d \delta x_D[k-i]$$

Ideal control effort (with d -step measurement delay):

$$\begin{aligned} \bar{\bar{f}}_{dx}[k] = a_x^{-1}[k] \Big\{ & \underbrace{\left[\Delta x_d[k] + (1 - \lambda_c) \sum_{i=1}^d \Delta x_d[k-i] \right]}_{\Delta x_d^d[k]} + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] \\ & - \underbrace{\left[\delta x_D[k] + (1 - \lambda_c) \sum_{i=1}^d \delta x_D[k-i] \right]}_{\delta x_D^d[k]} \Big\} \end{aligned}$$

Motion error:

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - \underbrace{\left\{ a_x[k]f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_x[k-i]f_T[k-i] \right\}}_{\delta x_T^d[k]}$$

Implementable control effort

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k] \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x_m[k] - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}}$$

Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k] [f_{dx}[k] - \bar{\bar{f}}_{dx}[k]] - \delta x_D^d[k]$$

$$a_x[k] \bar{\bar{f}}_{dx}[k] = \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] - \delta x_D^d[k]$$

$$a_x[k] f_{dx}[k] = (1 + e_{a_x}[k] \hat{a}_x^{-1}[k]) \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[(\delta x[k-d] + n_x[k]) - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}$$

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{\bar{f}}_{dx}[k] &= e_{a_x}[k] f_{dx}[k] + (1 - \lambda_c) n_x[k] \\ &\quad + (1 - \lambda_c) \sum_{i=1}^d (a_x[k-i] - \hat{a}_x[k-i]) f_{dx}[k-i] + \delta x_D^d[k] \end{aligned}$$

Parameter model

$$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]; \quad a_{x_{ram}}[k] \approx a'_x[k] x_{ram}[k]$$

$$\begin{aligned} a_x[k+1] &= a_{x_{det}}[k+1] + a_{x_{ram}}[k+1] = \underbrace{\{a_{x_{det}}[k] + a_{x_{ram}}[k]\}}_{a_x[k]} + \Delta a_x[k] + \underbrace{\{a_{x_{ram}}[k+1] - a_{x_{ram}}[k]\}}_{\Delta a_{ram}[k]} \\ &= a_x[k] + \Delta a_x[k] + \Delta a_{ram}[k] \end{aligned}$$

$$\Delta a_x[k] = a_{x_{det}}[k+1] - a_{x_{det}}[k] = a'_x[k] \Delta h_d[k] \text{ (from } p_d), \quad \Delta a_{ram}[k] = a'_x[k] \Delta x_{ram}[k]$$

$$\Delta h_d[k] = h_d[k+1] - h_d[k] \text{ (desired-trajectory height increment).}$$

Gain dynamics — $a_{x_{ram}}$ is first-order autoregressive (AR(1))

Assumption ($x = x_{det} + x_{ram}$, perfect tracking $x_{det} = x_d$): $\delta x = x_d - x = -x_{ram}$ (zero-mean), $x_{ram} = -\delta x$; ideal closed loop ($\hat{a}_x = a_x$, $\delta x_D^d \equiv 0$) gives $\delta x[k+1] = \lambda_c \delta x[k] - \varepsilon[k]$ (ε : closed-loop residual, defined in the boxed motion error below).

$$a_{x_{ram}}[k+1] = -a'_x \delta x[k+1] = -a'_x (\lambda_c \delta x[k] - \varepsilon[k]) = \lambda_c a_{x_{ram}}[k] + a'_x \varepsilon[k]$$

$$\begin{aligned} a_x[k+1] &= (a_{x_{det}}[k] + \Delta a_x[k]) + (\lambda_c a_{x_{ram}}[k] + a'_x \varepsilon[k]) \\ &= (a_{x_{det}}[k] + \Delta a_x[k]) + \lambda_c (a_x[k] - a_{x_{det}}[k]) + a'_x \varepsilon[k] \\ &= \underbrace{\lambda_c a_x[k]}_{\mathbf{F}_e(4,4)=\lambda_c} + \underbrace{(1 - \lambda_c) a_{x_{det}}[k] + \Delta a_x[k]}_{\text{known input}} + \underbrace{a'_x \varepsilon[k]}_{q_4[k]} \end{aligned}$$

$\mathbf{F}_e(4,4) = \lambda_c$

Parameter estimator (predict reverts to $a_{x_{det}}$): $\hat{a}_x[k+1] = \lambda_c \hat{a}_x[k] + (1 - \lambda_c) a_{x_{det}}[k] + \Delta a_x[k]$

Parameter estimation error: $e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$

Estimation error dynamics: $e_{a_x}[k+1] = \lambda_c e_{a_x}[k] + a'_x \varepsilon[k]$

$$a_x[k-i] - \hat{a}_x[k-i] = e_{a_x}[k-i] \stackrel{(\dagger)}{\approx} e_{a_x}[k] - \sum_{j=1}^i \Delta a_{ram}[k-j]$$

($\dagger$) feedforward form $e_{a_x}[k+1] = e_{a_x}[k] + \Delta a_{ram}$ is exact; the AR(1) predictor gives $e_{a_x}[k+1] = \lambda_c e_{a_x} + a'_x \varepsilon$, altering the expansion by $O((1-\lambda_c)e_{a_x})$ which enters only $\delta x_{\delta af}^d$ ($\ll \delta x_T^d$ in ε).

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{\bar{f}}_{dx}[k] &\approx e_{a_x}[k] \underbrace{\left\{ f_{dx}[k] + (1-\lambda_c) \sum_{i=1}^d f_{dx}[k-i] \right\}}_{F_{dx}[k]} \\ &\quad - \underbrace{\left\{ (1-\lambda_c) \sum_{i=1}^d (d+1-i) f_{dx}[k-i] \Delta a_{ram}[k-i] \right\}}_{\delta x_{\delta af}^d[k]} \\ &\quad + \delta x_D^d[k] + (1-\lambda_c) n_x[k] \end{aligned}$$

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] - \delta x_D^d[k] - \underbrace{\left\{ \delta x_T^d[k] - \delta x_{\delta af}^d[k] + (1-\lambda_c) n_x[k] \right\}}_{\varepsilon[k]}$$

State equation

($\delta x_D^d \equiv 0$: disturbance not modelled)

$$\delta x_1[k+1] = \delta x_2[k]$$

$$\delta x_2[k+1] = \delta x_3[k]$$

$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k]$$

$$a_x[k+1] = \lambda_c a_x[k] + (1-\lambda_c) a_{x_{det}}[k] + \Delta a_x[k] + a'_x \varepsilon[k]$$

Output equation

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} = a_x[k] - \sum_{i=1}^d \Delta a_x[k-i] + \underbrace{\left\{ n_a[k] - \sum_{i=1}^d \Delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{c}[k] = \begin{bmatrix} 0 \\ \sum_{i=1}^d \Delta a_x[k-i] \end{bmatrix} \text{ (known),} \quad \mathbf{y}[k] = \mathbf{H}\mathbf{x}[k] - \mathbf{c}[k] + \mathbf{r}[k]$$

Past-error expansion under AR(1) (†)

Two predictors, two e_{a_x} recursions (predict-only):

$$\text{feedforward: } e_{a_x}[k+1] = e_{a_x}[k] + \Delta a_{ram}[k] \Rightarrow e_{a_x}[k-i] = e_{a_x}[k] - \sum_{j=1}^i \Delta a_{ram}[k-j] \quad (\text{exact})$$

$$\text{AR(1): } e_{a_x}[k+1] = \lambda_c e_{a_x}[k] + a'_x \varepsilon[k] \Rightarrow e_{a_x}[k-i] = \lambda_c^{-i} (e_{a_x}[k] - \sum_{j=1}^i \lambda_c^{j-1} a'_x \varepsilon[k-j])$$

Difference ($\Delta_i := e_{a_x}^{\text{AR(1)}}[k-i] - e_{a_x}^{\text{ff}}[k-i]$):

$$\Delta_i = \underbrace{(\lambda_c^{-i} - 1) e_{a_x}[k]}_{\text{reverting}} - \lambda_c^{-i} \sum_{j=1}^i \lambda_c^{j-1} a'_x \varepsilon[k-j] + \sum_{j=1}^i \Delta a_{ram}[k-j]$$

Leading order (small i , $\lambda_c^{-i} \approx 1 + i(1 - \lambda_c)$):

$$(\lambda_c^{-i} - 1) e_{a_x}[k] \approx i(1 - \lambda_c) e_{a_x}[k] = O((1 - \lambda_c) e_{a_x})$$

$\Rightarrow$ the feedforward expansion errs by $O((1 - \lambda_c) e_{a_x})$, entering ε only through $\delta x_{\delta af}^d$ ($\ll \delta x_T^d$); numerically Q_{33}/R_{22} with $\text{Var}(\Delta a_{ram})$ vs q_4 leaves $\hat{a}_x$ unchanged.

Closed-loop Estimator

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta \hat{x}_1[k]$$

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = \delta x_m[k] - \delta \hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$\begin{aligned} e_{y_2}[k] &= y_2[k] - \hat{y}_2[k] = a_{xm}[k] - \left(\hat{a}_x[k] - \sum_{i=1}^d \Delta a_x[k-i] \right) \\ &= \left(a_x[k] - \sum_{i=1}^d \Delta a_x[k-i] + r_2[k] \right) - \hat{a}_x[k] + \sum_{i=1}^d \Delta a_x[k-i] \\ &= e_{a_x}[k] + \underbrace{\left\{ n_a[k] - \sum_{i=1}^d \Delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{aligned}$$

$$\delta \hat{x}_1[k+1] = \delta \hat{x}_2[k] + \ell_{11}[k] e_{y_1}[k] + \ell_{12}[k] e_{y_2}[k]$$

$$\delta \hat{x}_2[k+1] = \delta \hat{x}_3[k] + \ell_{21}[k] e_{y_1}[k] + \ell_{22}[k] e_{y_2}[k]$$

$$\delta \hat{x}_3[k+1] = \lambda_c \cdot \delta \hat{x}_3[k] + \ell_{31}[k] e_{y_1}[k] + \ell_{32}[k] e_{y_2}[k]$$

$$\hat{a}_x[k+1] = \lambda_c \hat{a}_x[k] + (1 - \lambda_c) a_{x_{det}}[k] + \Delta a_x[k] + \ell_{41}[k] e_{y_1}[k] + \ell_{42}[k] e_{y_2}[k]$$

Error dynamics

$$e_{\delta x_1}[k+1] = e_{\delta x_2}[k] - \ell_{11}[k] e_{\delta x_1}[k] - \ell_{12}[k] e_{a_x}[k]$$

$$e_{\delta x_2}[k+1] = e_{\delta x_3}[k] - \ell_{21}[k] e_{\delta x_1}[k] - \ell_{22}[k] e_{a_x}[k]$$

$$e_{\delta x_3}[k+1] = \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k] \cdot e_{a_x}[k] - \ell_{31}[k] e_{\delta x_1}[k] - \ell_{32}[k] e_{a_x}[k] - \underbrace{\varepsilon[k]}_{q_3[k]}$$

$$e_{a_x}[k+1] = \lambda_c e_{a_x}[k] - \ell_{41}[k] e_{\delta x_1}[k] - \ell_{42}[k] e_{a_x}[k] + \underbrace{a'_x \varepsilon[k]}_{q_4[k]}$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & \lambda_c \end{bmatrix}$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k - i]$$

$$\mathbf{e}[k + 1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H} \mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}[k] \mathbf{r}[k]$$

Process / measurement noise variances

Assumption ($x = x_{det} + x_{ram}$, perfect tracking): $x_{ram} = -\delta x$ (zero-mean), $a'_x := \frac{da_x}{dh} = -\frac{a_x K_h}{R}$, $K_h = \frac{1}{c} \frac{dc}{dh}$ (wall sensitivity).

Closed-loop tracking variance (thermal-driven, $\hat{a}_x = a_x$):

$$\text{Var}(x_{ram}) = \text{Var}(\delta x) = C_{\delta x} \sigma_{\delta h}^2, \quad C_{\delta x} = 2 + \frac{1}{1 - \lambda_c^2}, \quad \sigma_{\delta h}^2 := 4k_B T a_x$$

(i) **True increment** $\Delta a_{ram} = a'_x \Delta x_{ram}$ ($\rho_1 = \text{Cov}(x_{ram}[k+1], x_{ram}[k]) / \text{Var}(x_{ram})$, lag-1 autocorr):

$$\text{Var}(\Delta x_{ram}) = 2(1 - \rho_1) \text{Var}(x_{ram}), \quad \rho_1 = \lambda_c + \frac{2(1 - \lambda_c)}{C_{\delta x}} \Rightarrow 1 - \rho_1 = \frac{1}{(1 + \lambda_c) C_{\delta x}}$$

$$\boxed{\text{Var}(\Delta a_{ram}) = a_x'^2 \cdot \underbrace{2(1 - \rho_1) C_{\delta x}}_{= 2/(1 + \lambda_c)} \sigma_{\delta h}^2 = \frac{2}{1 + \lambda_c} a_x'^2 \sigma_{\delta h}^2 \quad (\rightarrow Q_{33}^{\text{randgain}}, R_{22}^{\text{delay}})}$$

(ii) **AR(1) white driver** $w_4 = a'_x \varepsilon$ ($\text{Var}(\varepsilon) = (1 - \lambda_c^2) \text{Var}(\delta x)$ since δx is AR(1)):

$$\boxed{q_4 = a_x'^2 \text{Var}(\varepsilon) = (1 - \lambda_c^2) C_{\delta x} a_x'^2 \sigma_{\delta h}^2 = (3 - 2\lambda_c^2) a_x'^2 \sigma_{\delta h}^2 \quad (\rightarrow Q_{44})}$$

Self-consistency (AR(1) steady-state variance = true gain variance):

$$\frac{q_4}{1 - \lambda_c^2} = C_{\delta x} a_x'^2 \sigma_{\delta h}^2 = \text{Var}(a_{x_{ram}}) \checkmark$$

Impl.: K_h , $\sigma_{\delta h}^2 = 4k_B T a_\perp$ evaluated at $\bar{h}_{\text{meas}}(p_m)$; gain $a_x \rightarrow \hat{a}_x$ (estimate); reversion target $a_{x_{det}}$ from p_d .

Original random-walk Q_{44} — the mismatch

Original model (RW, $\mathbf{F}_e(4,4) = 1$): predict mean and Q_{44} are mutually consistent,

$$\hat{a}_x[k+1] = \hat{a}_x[k] + \Delta a_x[k], \quad Q_{44} = \text{Var}(\Delta a_{ram}).$$

True increment (from $a_{x_{ram}}[k+1] = \lambda_c a_{x_{ram}}[k] + a'_x \varepsilon[k]$):

$$\Delta a_{ram}[k] = a_{x_{ram}}[k+1] - a_{x_{ram}}[k] = \underbrace{(\lambda_c - 1) a_{x_{ram}}[k]}_{\text{reversion (spring)}} + \underbrace{a'_x \varepsilon[k]}_{\text{white}}$$

Mismatch: RW puts the *whole* Δa_{ram} into Q_{44} — it treats the deterministic reversion $(\lambda_c - 1)a_{x_{ram}}$ as white noise; $\mathbf{F}_e(4,4) = 1$ keeps no spring, so the modelled gain variance is unbounded while the true gain is bounded:

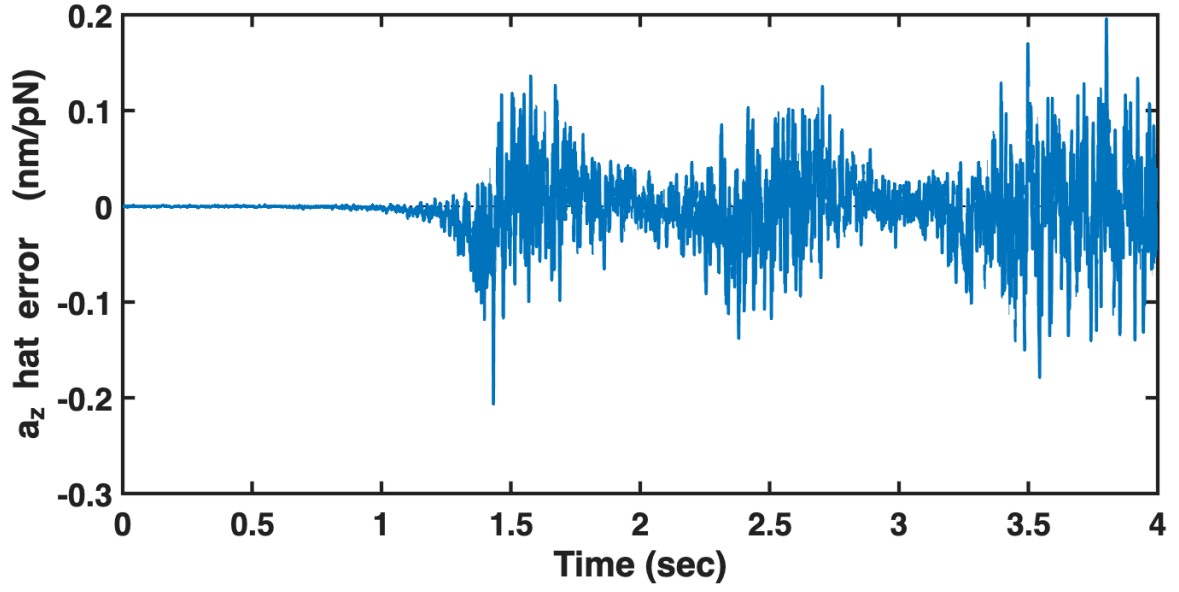
$$\text{RW: } P_a[k+1] = P_a[k] + Q_{44} \nearrow \infty \quad \text{AR(1): } P_a[k+1] = \lambda_c^2 P_a[k] + q_4 \rightarrow \frac{q_4}{1-\lambda_c^2}$$

Fix: move the reversion into the dynamics ($\mathbf{F}_e(4,4) = \lambda_c$, predict reverts to $a_{x_{det}}$), keep only the white driver in Q_{44} :

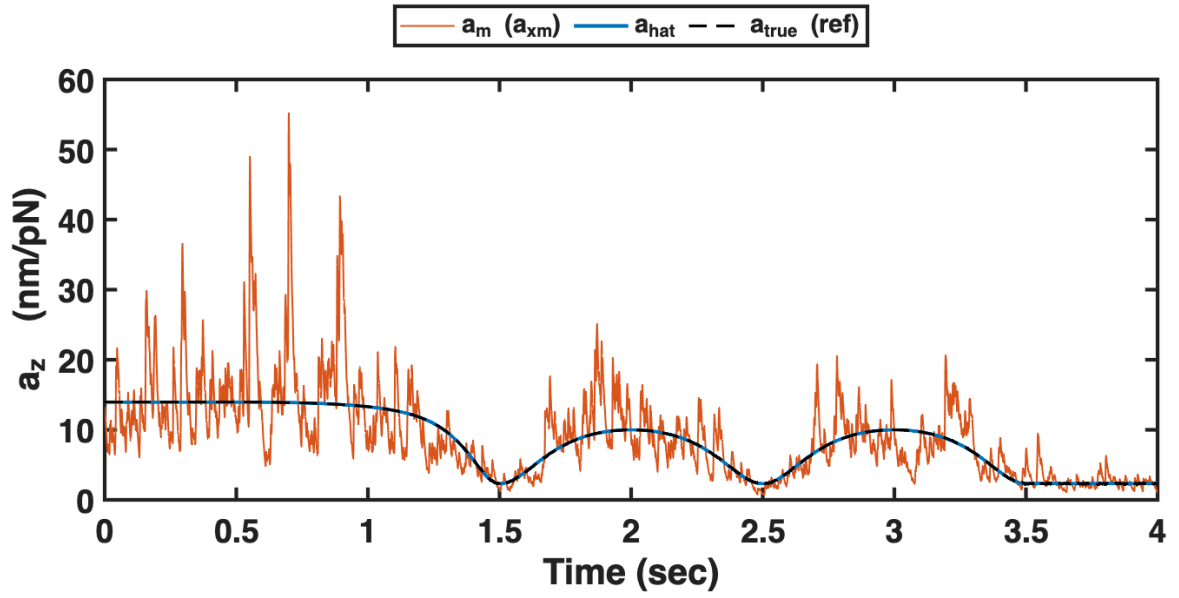
$$\underbrace{\text{Var}(\Delta a_{ram}) = \frac{2}{1+\lambda_c} a_x'^2 \sigma_{\delta h}^2}_{\text{RW } Q_{44}} \longrightarrow \underbrace{q_4 = (3 - 2\lambda_c^2) a_x'^2 \sigma_{\delta h}^2}_{\text{AR(1) } Q_{44}}$$

Effect (1 Hz, near wall, z -axis): $P_a/\text{Var}(a_{x_{ram}}) : 19.67 \rightarrow 0.78, \quad |\hat{a}_z - a_z| : 9.4\% \rightarrow 0.4\%.$

Single-seed time series (1 Hz, near wall, z -axis)



a_z estimation error $\hat{a}_z - a_z$.



a_z : measurement a_m , EKF estimate $\hat{a}_x$, truth a_{true} .

Decoupling from the known wall model — $a_{x_{det}}$, a'_x from model-free sources

Both $a_{x_{det}}[k]$ (reversion target) and $a'_x[k]$ (feedforward slope, via $\Delta a_x[k] = a'_x[k]\Delta h_d[k]$) above are read from the *known* wall model $c(\bar{h})$. For an unknown wall, replace them with model-free, exogenous quantities — the state stays 4-dimensional, no augmented slope/level state is added:

$$\hat{a}_{x_{det}}[k+1] = (1 - \alpha) \hat{a}_{x_{det}}[k] + \alpha a_{xm}[k+1] \quad (\text{EWMA of } a_{xm}; \text{ needs only } \lambda_c, 4k_B T, \text{ not } c(\bar{h}))$$

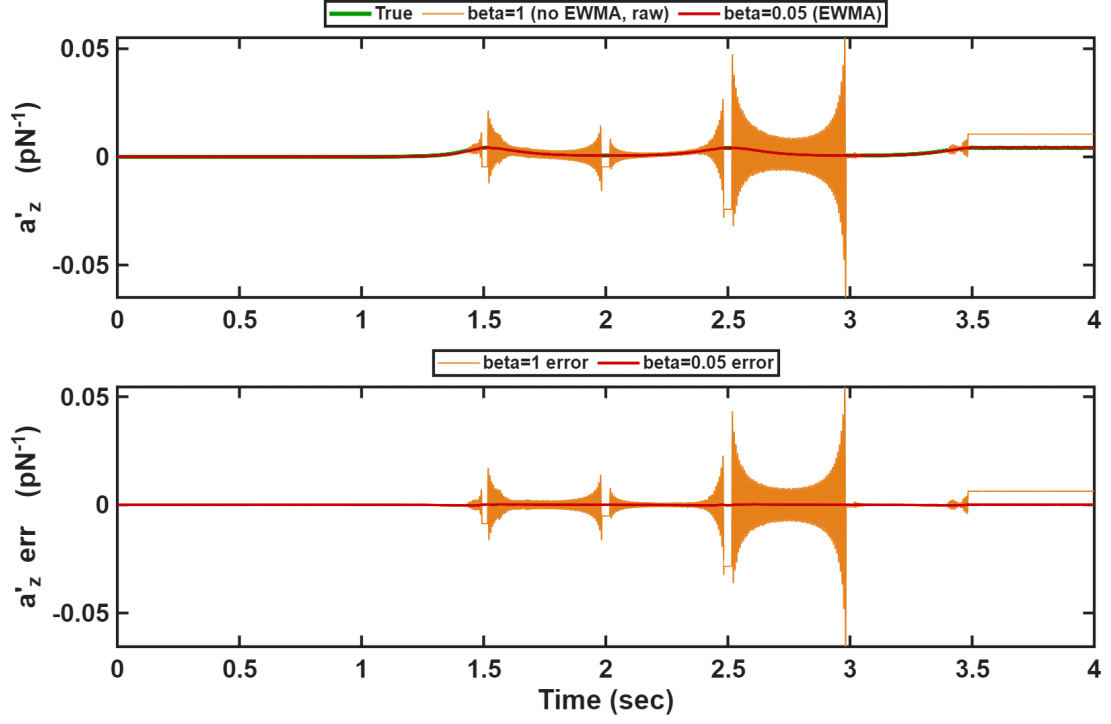
$$\hat{a}'_x[k] = \begin{cases} (1 - \beta) \hat{a}'_x[k-1] + \beta \frac{\hat{a}_x[k] - \hat{a}_x[k-1]}{\Delta h_d[k-1]}, & |\Delta h_d[k-1]| > \gamma \\ \hat{a}'_x[k-1], & |\Delta h_d[k-1]| \leq \gamma \text{ (gate closed, frozen)} \end{cases}$$

γ (gate) avoids dividing by a near-zero Δh_d (e.g. during a height hold); β (EWMA) trades noise against lag. Test values: $\gamma = 10^{-3} \mu\text{m}$, $\beta = 0.05$. Define the two new estimation errors:

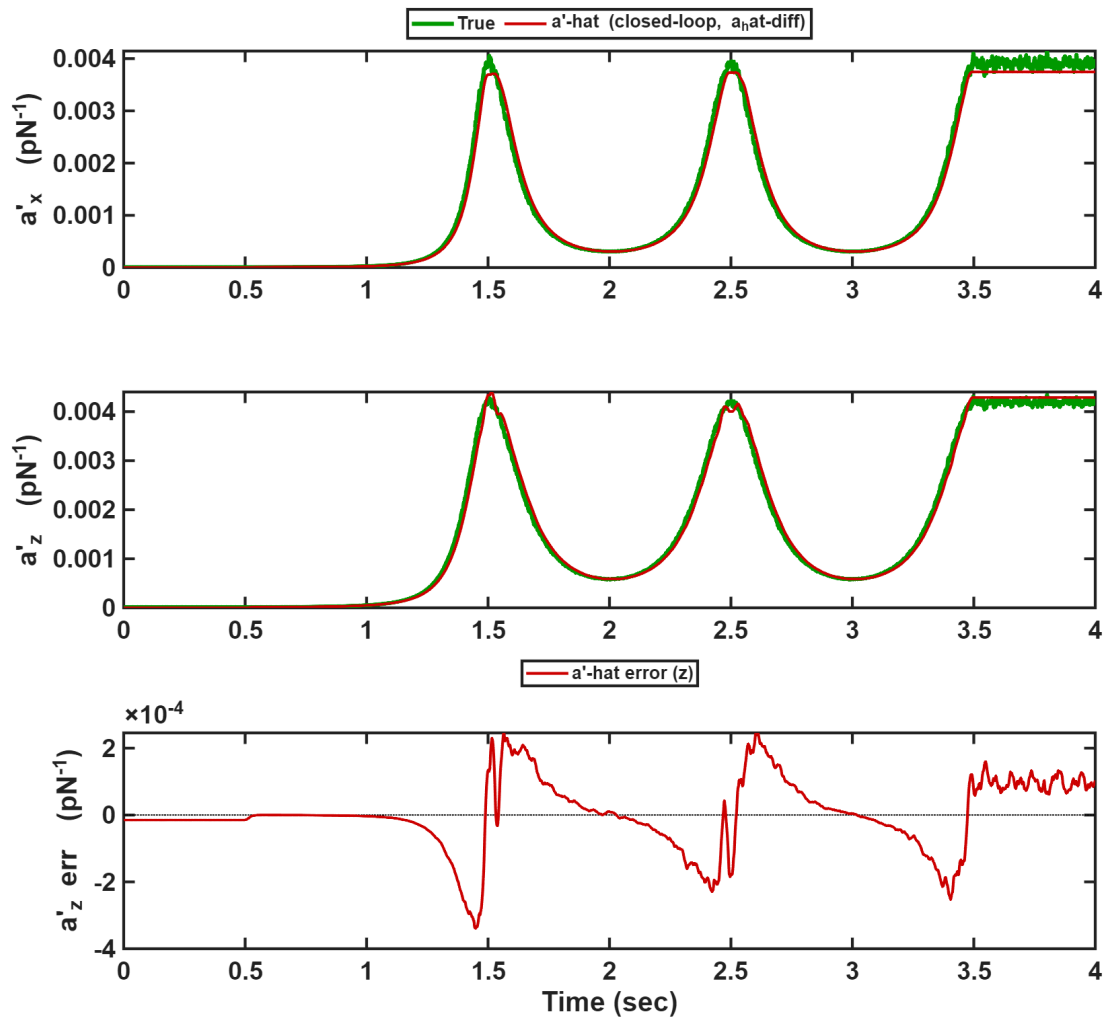
$$e_{a_{x_{det}}}[k] := a_{x_{det}}[k] - \hat{a}_{x_{det}}[k], \quad e_{a'_x}[k] := a'_x[k] - \hat{a}'_x[k]$$

Verification — osc_1hz, near wall (three closed-loop tests)

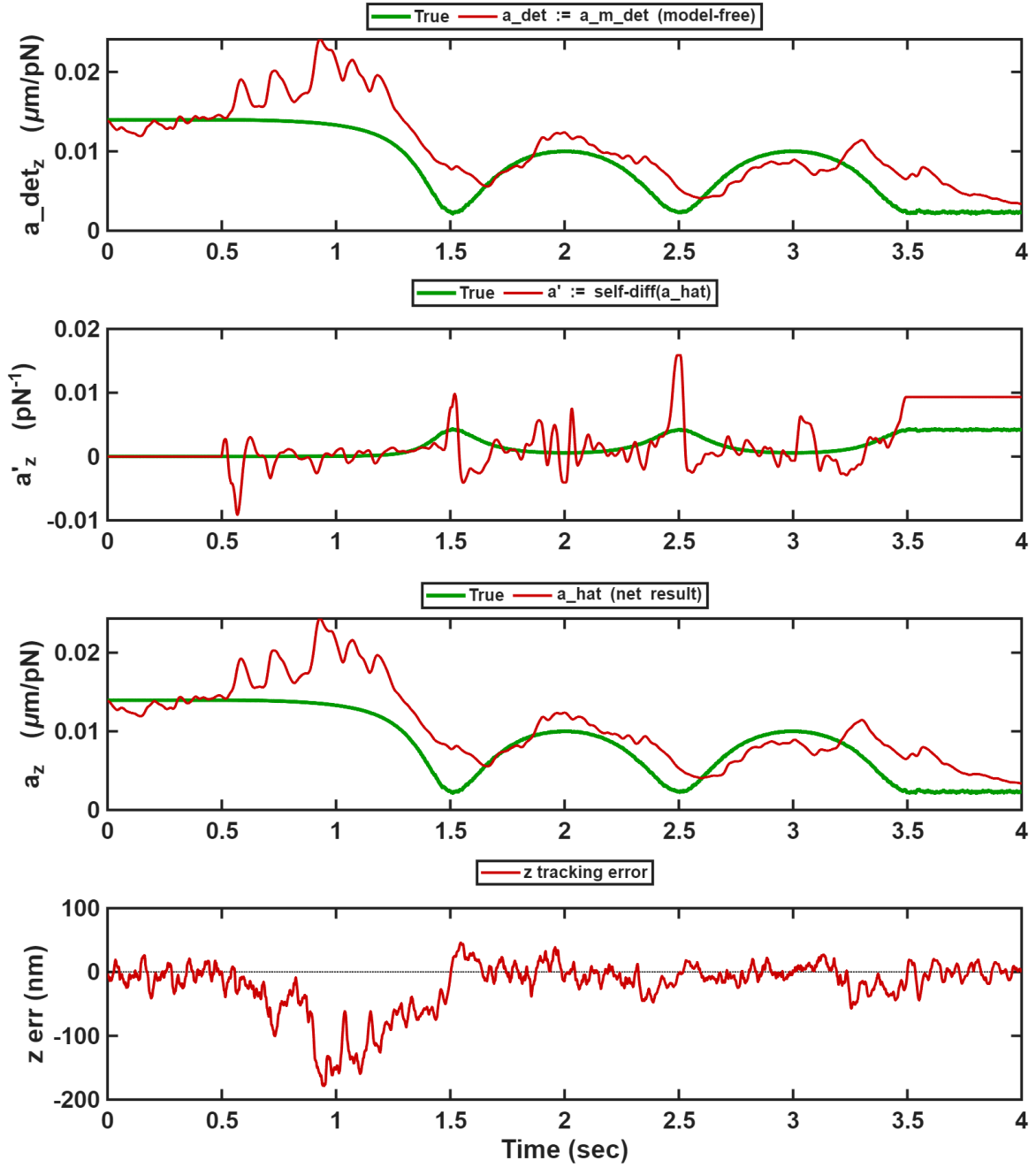
Same scenario as the AR(1) verification above ($h_{init} = 50$, trough $\bar{h} \approx 1.2$, $\lambda_c = 0.7$).



Layer 1 — self-differenced a'_z , raw ($\beta = 1$) vs EWMA-smoothed ($\beta = 0.05$): $corr_z$ 0.36 $\rightarrow$ 0.997. Smoothing alone does most of the work.



Layer 2 — same smoothed estimate, full closed loop, $\hat{a}_{x_{\text{det}}}$ still known-wall: $\text{corr} = 0.99$, but circular ($\hat{a}_x \approx a_{x_{\text{det}}}$, riding on the smoothing above).



Layer 3 — both $\hat{a}_{x_{det}}$ and $\hat{a}'_x$ model-free, stable (0/3 diverge): $a_{x_{det}}$ corr= 0.70; a'_x corr= 0.27 (weak link, honest number).

Corrected error dynamics

The estimator predict becomes $\hat{a}_x[k+1] = \lambda_c \hat{a}_x[k] + (1 - \lambda_c) \hat{a}_{x_{det}}[k] + \hat{a}'_x[k] \Delta h_d[k]$. Subtracting from the true dynamics ($a_x[k+1] = \lambda_c a_x[k] + (1 - \lambda_c) a_{x_{det}}[k] + \Delta a_x[k] + a'_x \varepsilon[k]$, boxed above):

$$e_{a_x}[k+1] = \lambda_c e_{a_x}[k] + (1 - \lambda_c) e_{a_{x_{det}}}[k] + e_{a'_x}[k] \Delta h_d[k] + a'_x \varepsilon[k]$$

Two new noise terms appear (boxed); the original term $a'_x \varepsilon[k]$ is unchanged. $\mathbf{F}_e$ and $\mathbf{H}$ keep their original $4 \times 4/2 \times 4$ form — $\hat{a}_{x_{det}}$ and $\hat{a}'_x$ enter as exogenous inputs, exactly like $a_{x_{det}}$ and a'_x did; only Q_{44} , R_{22} change.

Corrected Q_{44} , R_{22}

The output equation's known correction $\sum_{i=1}^d \Delta a_x[k-i] = a_{x_{det}}[k] - a_{x_{det}}[k-d] \approx a'_x[k] \Delta H_d[k]$

($\Delta H_d[k] := h_d[k] - h_d[k-d]$, small- d telescoping approx., already implicit in the parameter model) is now computed as $\hat{a}'_x[k] \Delta H_d[k]$, feeding the same $e_{a'_x}$ into the measurement side:

$$\boxed{\begin{aligned} Q_{44}^{\text{new}}[k] &= \underbrace{(3 - 2\lambda_c^2) a_x^2 \sigma_{\delta h}^2}_{q_4, \text{ original}} + (1 - \lambda_c)^2 \text{Var}(e_{a_{x_{det}}})[k] + \Delta h_d[k]^2 \text{Var}(e_{a'_x})[k] \\ R_{22}^{\text{new}}[k] &= \underbrace{R_{22}^{\text{intrinsic}} + R_{22}^{\text{delay}}}_{\text{original}} + \Delta H_d[k]^2 \text{Var}(e_{a'_x})[k] \end{aligned}}$$

Assumption: the three driving noises ($a'_x \varepsilon$, $e_{a_{x_{det}}}$, $e_{a'_x}$) are mutually independent – revisit if not.

$\text{Var}(e_{a_{x_{det}}})$ — closed form

$\hat{a}_{x_{det}}$ is a first-moment EWMA of $a_{xm}[k] = a_x[k-d] + n_a[k]$ (n_a : variance = R_{22} , the a_{xm} measurement noise derived in the companion a_{xm} documents). Standard EWMA group delay $\tau_\alpha = (1-\alpha)/\alpha$ samples, plus the EWMA noise-gain $\alpha/(2-\alpha)$ (same form as K_{var} 's g_E factor elsewhere in this line):

$$\boxed{\text{Var}(e_{a_{x_{det}}})[k] \approx \underbrace{\left(\Delta a_x[k] \cdot \frac{1-\alpha}{\alpha} \right)^2}_{\text{EWMA lag bias}} + \underbrace{\frac{\alpha}{2-\alpha} R_{22}[k]}_{\text{noise averaging}}}$$

Impl.: $\alpha = \mathbf{a_det_lp}$ (currently 0.005, inherited from an unrelated, now-superseded role – *never* tuned as the AR(1) reversion target's own bandwidth). At $\alpha = 0.005$, $\tau_\alpha \approx 199$ samples; the lag term is suspected to dominate for a fast (1 Hz) trajectory. A sweep of α trades the lag term against the noise term – the same lag $\leftrightarrow$ noise trade-off as the Q_{55} κ -sweep in the companion `5state_est_aprime` line.

$\text{Var}(e_{a'_x})$ — self-referential, closed form under two stated approximations

$\hat{a}'_x[k]$ is differenced from $\hat{a}_x$'s own history, so $e_{a'_x}[k]$ is a function of $e_{a_x}[k] - e_{a_x}[k-1]$ – *self-referential* with the very quantity Q_{44}^{new} is meant to bound (same structural family as the $a_{x_{det}} := \hat{a}_x$ closed-loop instability, Exp. 03 in the companion status note, but here at the level of noise statistics rather than outright divergence). A closed form IS derivable, under two named approximations: **(A)** Δh_d locally constant over the EWMA window; **(B)** e_{a_x} 's driving noise treated as white when computing e_{a_x} 's own autocorrelation (ignores that $e_{a_{x_{det}}}$, $e_{a'_x}$ are themselves EWMA outputs and hence colored – the same order of simplification as the Q33 “Path C” treatment elsewhere in this project line).

Step 1 — unroll the EWMA. With $r[k] := [\hat{a}_x[k] - \hat{a}_x[k-1]]/\Delta h_d[k-1]$ and $d[k] := e_{a_x}[k] - e_{a_x}[k-1]$ (so $a'_x[k] - r[k] = d[k]/\Delta h_d[k-1]$, boxed earlier):

$$\hat{a}'_x[k] = \beta \sum_{j \geq 0} (1-\beta)^j r[k-j] \Rightarrow e_{a'_x}[k] \approx \frac{\beta}{\Delta h_d[k-1]} \sum_{j \geq 0} (1-\beta)^j d[k-j]$$

Step 2 — model e_{a_x} as AR(1), pole λ_c (approximation B). Standard AR(1) increment/autocorrelation identities (same form as $\text{Var}(\Delta x_{ram})$ earlier):

$$\text{Var}(d) = 2(1 - \lambda_c) P_a, \quad \rho_d(\tau) = -\frac{1-\lambda_c}{2} \lambda_c^{\tau-1} \quad (\tau \geq 1)$$

Step 3 — EWMA variance transfer (linear, no Isserlis needed – d itself, not d^2 , is being averaged): the color sum telescopes exactly,

$$1 + 2 \sum_{\tau \geq 1} \rho_d(\tau) s^\tau = \frac{\beta}{1 - \lambda_c(1-\beta)}, \quad s := 1 - \beta$$

$$\text{Var}(e_{a'_x}) \approx \frac{2\beta^2(1-\lambda_c)}{(2-\beta)[1-\lambda_c(1-\beta)]} \cdot \frac{P_a}{\Delta h_d[k-1]^2}$$

Step 4 — self-consistency (substitute into Q_{44}^{new} , $\Delta h_d[k] \approx \Delta h_d[k-1]$, impose $P_a = Q_{44}^{\text{new}}/(1-\lambda_c^2)$):

$$P_a = \frac{q_4 + (1-\lambda_c)^2 \text{Var}(e_{a_{x_{det}}})}{(1-\lambda_c) \left[(1+\lambda_c) - \frac{2\beta^2}{(2-\beta)[1-\lambda_c(1-\beta)]} \right]}$$

$\text{Var}(e_{a'_x})$ then follows by plugging this P_a back into the Step 3 box.

Stability condition: $P_a \geq 0$ requires the bracket > 0 , i.e. $(1+\lambda_c) > \frac{2\beta^2}{(2-\beta)[1-\lambda_c(1-\beta)]}$ – a closed-form bound on how little smoothing (β) the self-differenced a' can tolerate before the self-consistent variance has no valid solution. **Check** ($\lambda_c = 0.7$): $\beta = 0.05 \Rightarrow \text{bracket} \approx 1.69 > 0$ (stable, matches the verified test config); $\beta = 1$ (raw, no smoothing) $\Rightarrow \text{bracket} = -0.3 < 0$ – no valid P_a , consistent with the empirically observed instability of un-smoothed self-differencing.

Caveat: this is a closed form *under approximations (A) and (B)* – not an exact result. (B) in particular ignores that $e_{a_{x_{det}}}$ and $e_{a'_x}$ are themselves colored (EWMA outputs), so the true $\rho_d(\tau)$ has additional structure beyond the pure-AR(1) form used here; treat the stability boundary as indicative, not exact, and prefer the closed-loop empirical test (Exp. noheat above) as the final check.

FALSEIFICATION — $e_{a_{x_{det}}}$ is NOT exogenous in closed loop

The corrected Q_{44}^{new} above was implemented and tested on the failing scenario: **no effect** (bias $+11.9\% \rightarrow +12.2\%$). Loop-cut interventions on the same scenario: (i) control law fed the *true* gain (estimator untouched) $\Rightarrow$ fully stable, open-loop anchor bias only 1.6–3.7%; (ii) y_2 KF channel permanently off $\Rightarrow$ blow-up *bit-identical* to the closed-loop run. Hence the destabilizing coupling enters through the *real tracking error* into the predict-side anchor ($\hat{a}_{x_{det}} \rightarrow$ predict, fixed weight $1 - \lambda_c$, no Kalman damping) — a path that no Q/R re-weighting can see. The exogeneity assumption behind Q_{44}^{new} is falsified; the sections below replace it with the explicit coupling.

The g -mismatch closed loop

Let $g[k] := a_x[k]/\hat{a}_x[k]$ (gain mismatch; control uses $\hat{a}_x$). Re-deriving the closed loop from the implementable control effort (boxed near the top of this document) *without* assuming $\hat{a}_x = a_x$, quasi-static g over the $d+1$ window:

$$\boxed{\delta x[k+1] = \lambda_c \delta x[k] + (1-g)(1-\lambda_c) \delta x[k-d] + (1-g) D[k] - \varepsilon_g[k]}$$

$$D[k] := x_d[k+1] - \lambda_c x_d[k] - (1-\lambda_c) x_d[k-d]$$

$$\varepsilon_g[k] := a_x f_T[k] + (1-\lambda_c) \sum_{i=1}^d a_x f_T[k-i] - g(1-\lambda_c) n_x[k]$$

$D[k]$ is a *known* trajectory functional ($= 0$ during holds). Self-check $g = 1$: reduces to $\lambda_c \delta x - \varepsilon$ with $\text{Var}(\varepsilon) = 4k_B T a_x (1 + (1-\lambda_c)^2 d) + (1-\lambda_c)^2 \sigma_{n_x}^2$ — identical to Q_{33} at $f_d = 0$. **Validation** (fixed- g probe, noise-free osc): corr = 1.0000, amplitude ratio 1.000, rel. RMS $\leq 0.3\%$ at $g = 0.5$ and $g = 2$.

δp_{md} **is the hidden g channel**: the deterministic response is, quasi-statically,

$$\delta x_{det}[k] \approx \frac{(1-g) D[k]}{g(1-\lambda_c)}$$

(DC gain of the loop above), and this is precisely the content of the LP output δp_{md} — a *signed, first-moment* measurement of g that the estimator chain currently discards.

Variance map under mismatch

Augmenting $[\delta x[k], \delta x[k-1], \delta x[k-2], w[k-1], w[k-2], b[k-1]]$ ($w := a_x f_T$, b the a_{pd} -LP state) and solving the discrete Lyapunov equation gives the HP-residual variance as a g -family:

$$\boxed{\text{Var}(\delta x_r)(g) = c_w(g) \cdot 4k_B T a_x + c_n(g) \cdot \sigma_{n_x}^2}$$

with the consistency check $c_w(1) \cdot 4k_B T a_x + c_n(1) \sigma_{n_x}^2 \equiv C_{\text{dpmr}} 4k_B T a_x + C_n \sigma_{n_x}^2$. **Validation** (hold at $\bar{h} = 2.22$, five fixed g): predicted vs measured within 0.1–0.5%; the $g = 1$ cross-check reproduces the C_{dpmr} formula to ratio 1.0000. The confound in numbers: $a_{xm}/a_x = 1.22$ at $g = 0.6$, 0.93 at $g = 1.67$ — overestimating $\hat{a}_x$ inflates a_{xm} , which (fed back as the anchor) pushes $\hat{a}_x$ further up: positive feedback with loop strength $\propto \alpha$. At hold the DC loop gain is $\approx 0.31 < 1$ (statically stable), so the cliff requires trajectory motion plus noise.

Mechanism closure — bias and the α cliff

A scalar surrogate built from ONLY the two boxed results above (no EKF, no y_1/y_2 , no 3-axis physics):

- *Mean path*: reproduces the closed-loop bias (26.1%/10.6% predicted vs 25.5%/11.9% measured at $\alpha = 0.002/0.005$) but stays stable to $\alpha = 0.05 \Rightarrow$ the bias is a mean-path (lag + loop-equilibrium) effect, the cliff is not.
- *With correlated noise sampling*: cliff appears at the measured location ($0.007 \rightarrow 0/5$, $0.008 \rightarrow 1/5$, $0.010 \rightarrow 3/5$, $0.020 \rightarrow 5/5$; real simulator: 0.005 stable, 0.008 blown) $\Rightarrow$ the cliff is noise-triggered escape (χ^2 fluctuation of $\hat{\sigma}_{\delta_{x_r}}^2$, rel. std $\sim 46\%$, amplified through the nonlinear g -loop).
- *Counter-side (oracle)*: restoring the missing g -dependence inside the fully closed loop (true g , correction below) collapses the bias to +4.1% and removes the cliff ($0.008, 0.010 \rightarrow 0/3$).

Deployable $\hat{g}$ estimator from δp_{md} (model-free)

Apply the SAME LP and d -step delay to the known regressor: $\bar{D}[k] := \text{LP}_{a_{pd}}(D[k-d])$, so that lags match:

$$\mathbb{E}\{\delta p_{md}[k]\} \approx \frac{1-g}{g(1-\lambda_c)} \bar{D}[k]$$

The pointwise ratio $u = (1-\lambda_c) \delta p_{md} / \bar{D}$ FAILS in practice (per-sample SNR < 1 ; ratio rectification — tested: bias +52.9%, worse than uncorrected). The **regression form** is required — excitation-weighted, and unbiased because $\bar{D}$ is deterministic ($\mathbb{E}\{\bar{D} \cdot \text{noise}\} = 0$):

$$S_{xy} = \text{EWMA}_{\beta_g}(\bar{D} \delta p_{md}), \quad S_{xx} = \text{EWMA}_{\beta_g}(\bar{D}^2), \quad u = (1-\lambda_c) \frac{S_{xy}}{S_{xx}}, \quad \hat{g} = \text{clamp}\left(\frac{1}{1+u}, [0.5, 2]\right)$$

$\hat{g}$ held at its previous value while $S_{xx} \leq \gamma_D^2$ (insufficient excitation, e.g. holds). Test values: $\beta_g = 0.05$, $\gamma_D = 2 \times 10^{-3} \mu\text{m}$. (Slower $\beta_g = 0.02$ was tested and rejected: estimator lag destabilizes at higher α .)

Corrected inversion and anchor (deployable)

Run the deterministic-leak predictor with $\hat{g}$ (same recursion as the boxed g -mismatch loop, driven by the known $D[k]$), pass it through the same LP/HP as the measurement chain, and restore the g -dependence of the inversion:

$$\delta x_{det}[k+1] = \lambda_c \delta x_{det}[k] + (1-\hat{g})(1-\lambda_c) \delta x_{det}[k-d] + (1-\hat{g}) D[k]$$

$$\hat{\sigma}_{det}^2 = \text{EWMA}_{a_{cov}}(\text{HP}_{a_{pd}}(\delta x_{det}[k-d])^2)$$

$$a_{xm}^{corr}[k] = \frac{\hat{\sigma}_{\delta_{x_r}}^2[k] - \hat{\sigma}_{det}^2[k] - c_n(\hat{g}) \sigma_{n_x}^2}{c_w(\hat{g}) \cdot 4k_B T}, \quad \hat{a}_{x_{det}}[k+1] = (1-\alpha) \hat{a}_{x_{det}}[k] + \alpha a_{xm}^{corr}[k]$$

Only the anchor consumes a_{xm}^{corr} ; the y_2 channel keeps the raw a_{xm} (proven irrelevant by the loop-cut). $c_w(\cdot), c_n(\cdot)$ from a lookup over $\log g$ built offline (depends only on λ_c, a_{pd} — no $c(\bar{h})$).

Complete controller (unknown $c(\bar{h})$, wall position known)

Everything else in this document is unchanged. The full per-axis loop:

1. Control effort: boxed implementable form (top of document), $\hat{a}_x$ from the EKF.
2. EKF: 4-state AR(1), $F_e/H/Q_{33}/Q_{44}/R$ unchanged; predict $\hat{a}_x[k+1] = \lambda_c \hat{a}_x[k] + (1-\lambda_c) \hat{a}_{x_{det}}[k] + \hat{a}_x'[k] \Delta h_d[k]$.
3. $\hat{a}_x'$: gated+EWMA self-difference of $\hat{a}_x$ (boxed earlier; zero-cost, ablation-verified).

4. $\hat{g}$: regression estimator from δp_{md} (boxed above).

5. a_{xm}^{corr} , $\hat{a}_{x_{det}}$: corrected inversion and anchor (boxed above).

Residual known-model dependencies, declared out of scope for this line: K_h inside the Q_{44}/R_{22} random-gain floor, and the wall-aware $\hat{a}_x[0]$ seed.

End-to-end validation (osc_1hz, near wall, z-axis)

variant	α	blown	track _z (nm)	bias _z (%)
known wall (baseline)	–	0/3	24.1	0.0
model-free, uncorrected	0.005	0/3	34.9	+12.2
model-free, uncorrected	0.008	–	blown (> 1500 nm)	
oracle g (upper bound)	0.005	0/3	27.9	+4.1
oracle g (upper bound)	0.010	0/3	28.8	+1.6
deployable ($\hat{g}$ from δp_{md})	0.005	0/5	30.5	+3.2
deployable	0.008	1/5	34.0	+0.4
deployable	0.010	1/5	47.6	+0.1
deployable	0.020	5/5	blown	

Operating point: $\alpha = 0.005$ (0/5 seeds, all axes 28–31 nm, gain bias $\leq 5.5\%$ all axes), safety margin to the cliff $\approx 1.6\times$. The deployable controller recovers most of the oracle’s gain: bias $+12.2\% \rightarrow +3.2\%$ at the cost of ~ 6 nm tracking vs the known-wall baseline.